



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2025 Year 12 Course Assessment Task 4 (Trial Examination)

Friday, 1 August 2025

General instructions

- Working time – 3 hours.
(plus 10 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided (on page 17)

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: # BOOKLETS USED:

Class (please ✓)

☐ 12MXX.1 – Ms Ham

☐ 12MXX.2 – Mr Ho

Marker's use only.

QUESTION	1-10	11	12	13	14	15	16	%
MARKS	$\overline{10}$	$\overline{17}$	$\overline{15}$	$\overline{14}$	$\overline{14}$	$\overline{16}$	$\overline{14}$	$\overline{100}$

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer grid provided (labelled as page 17).

Glossary

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ – set of all natural numbers
- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3\}$ – set of all integers.
- $\mathbb{Z}^+$ – set of all positive integers (excludes zero)
- $\mathbb{R}$ – set of all real numbers
- $\mathbb{C}$ – set of all complex numbers

Questions

Marks

1. Let $\overrightarrow{OP}$, $\overrightarrow{OQ}$ and $\overrightarrow{OR}$ be the position vectors of points P , Q , R respectively. It is given that 1

$$\overrightarrow{PQ} = \begin{pmatrix} -4 \\ 1 \\ 8 \end{pmatrix}, \quad \overrightarrow{QR} = \begin{pmatrix} \lambda + 2 \\ 2 \\ \mu - 5 \end{pmatrix}$$

What are the values of λ and μ if P , Q and R are collinear?

- (A) $\lambda = -4$ and $\mu = 9$ (C) $\lambda = -6$ and $\mu = 13$
 (B) $\lambda = -6$ and $\mu = 11$ (D) $\lambda = -10$ and $\mu = 21$

2. Consider the two lines with vector equations 1

$$\underline{r}_1 = \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} \quad \text{and} \quad \underline{r}_2 = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -4 \\ -2 \end{pmatrix}$$

Which of the following is the exact value of the angle between the two lines?

- (A) $\cos^{-1} \left(\frac{7}{\sqrt{174}} \right)$ (C) $\cos^{-1} \left(\frac{7}{\sqrt{294}} \right)$
 (B) $\cos^{-1} \left(\frac{-13}{\sqrt{174}} \right)$ (D) $\cos^{-1} \left(\frac{-13}{\sqrt{294}} \right)$

3. Consider $z^6 = 1$. Which of the following are the roots of the equation? 1

- (A) $e^{i\frac{k\pi}{3}}$ for $k = \pm 1, \pm 2, \pm 3$ (C) $e^{i\frac{k\pi}{3}}$ for $k = 0, 1, 2, \dots, 5$
 (B) $e^{i\frac{k\pi}{6}}$ for $k = \pm 1, \pm 2, \pm 3$ (D) $e^{i\frac{k\pi}{6}}$ for $k = 0, 1, 2, \dots, 5$

4. Consider the statement: 1

$$\forall x \in \mathbb{R}, \quad \exists y \in \mathbb{R} \quad \text{such that} \quad x^y > 1$$

Which of the following is the negation of the statement above?

- (A) $\forall x \in \mathbb{R}, \quad \exists x \in \mathbb{R}, \quad x^y \leq 1$ (C) $\forall x, y \in \mathbb{R}, \quad x^y \leq 1$
 (B) $\exists x \in \mathbb{R}, \quad \forall y \in \mathbb{R}, \quad x^y \leq 1$ (D) $\exists x, y \in \mathbb{R}, \quad x^y \leq 1$

5. A particle moving along a horizontal line has acceleration 1

$$a = \frac{12 - 4x}{x^3}$$

where x is the displacement of the particle. The particle starts from rest at $x = 6$.

Which of the following describes the velocity of the particle?

- (A) $v^2 = -\frac{6}{x^2} + \frac{4}{x}$ (C) $v^2 = -\frac{6}{x^2} + \frac{4}{x} - \frac{1}{2}$
 (B) $v^2 = -\frac{12}{x^2} + \frac{8}{x}$ (D) $v^2 = -\frac{12}{x^2} + \frac{8}{x} - 1$

6. Let ω be a non-real cube root of unity. 1

Which of the following is equivalent to $(1 + 2\omega - \omega^2 + \omega^3)^5$?

- (A) -243ω (C) 243ω
 (B) $-243\omega^2$ (D) $243\omega^2$

Examination continues overleaf...

7. Let z be a complex number such that $\text{Im}(z) \neq 0$.

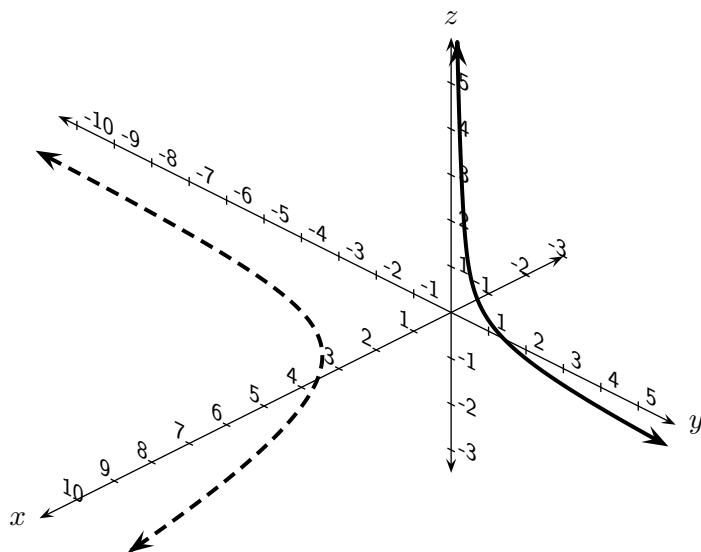
1

Which of the following expressions is equivalent to $\frac{8\bar{z}z}{(\bar{z}-z)^2}$?

- (A) $2 + 2\left(\frac{\text{Re}(z)}{\text{Im}(z)}\right)^2$ (C) $-2 - 2\left(\frac{\text{Re}(z)}{\text{Im}(z)}\right)^2$
 (B) $2 + 2\left(\frac{\text{Im}(z)}{\text{Re}(z)}\right)^2$ (D) $-2 - 2\left(\frac{\text{Im}(z)}{\text{Re}(z)}\right)^2$

8. Which position vector best describes the curve in diagram?

1



- (A) $\underline{r} = \begin{pmatrix} t \\ -\frac{1}{t} \\ e^{-t} \end{pmatrix}$ (C) $\underline{r} = \begin{pmatrix} e^{-t} \\ -\frac{1}{t} \\ t \end{pmatrix}$
 (B) $\underline{r} = \begin{pmatrix} t \\ \frac{1}{t} \\ e^t \end{pmatrix}$ (D) $\underline{r} = \begin{pmatrix} e^t \\ -\frac{1}{t} \\ -t \end{pmatrix}$

9. Which of the following integrals is equivalent to

1

$$\int_0^{\frac{\pi}{2}} \frac{x}{1 + \sin x + \cos x} dx$$

(A) $\frac{\pi}{2} \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin x - \cos x} dx$

(C) $\frac{\pi}{2} \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin x + \cos x} dx$

(B) $\frac{\pi}{4} \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin x - \cos x} dx$

(D) $\frac{\pi}{4} \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin x + \cos x} dx$

10. Consider the statement:

1

Period 0 is cancelled if and only if there is an excursion.

Which of the following statements is FALSE?

- (A) The negation is an ‘or’ statement.
- (B) The negation is ‘Period 0 being cancelled does not imply that there is an excursion’.
- (C) A counterexample is ‘There is no excursion and Period 0 is cancelled’.
- (D) A counterexample is ‘Period 0 is not cancelled and there is an excursion’.

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (17 marks)

Marks

(a) Find $\frac{-3+i}{1-2i}$, showing all working out. **2**

(b) Given that $w = 2 - i(2\sqrt{3})$ and $\text{Arg}(zw) = \frac{\pi}{3}$, find the exact value of $\text{Arg}(z)$. **2**

(c) The complex numbers $z_1 = 2 - 3i$, z_2 and z_3 are roots of the polynomial **2**

$$p(z) = z^3 - 8z^2 + 29z - 52$$

Find z_2 and z_3 .

(d) Find the square roots of $21 - 20i$. **2**

(e) Prove that $\frac{3}{2}k^2 - 6k - \frac{63}{2}$ is divisible by 12 for all positive odd integer k . **2**

(f) i. By decomposing $\frac{9x}{(x^2+2)(x+1)}$ into partial fractions, show that **2**

$$\frac{9x}{(x^2+2)(x+1)} = \frac{3x+6}{x^2+2} - \frac{3}{x+1}$$

ii. Hence find $\int \frac{9x}{(x^2+2)(x+1)} dx$. **2**

(g) Using an appropriate substitution, find $\int \frac{2}{1+\sqrt{2x}} dx$. **3**

Question 12 (15 marks)**Marks**

- (a) Consider the statement:

For all irrational numbers p , the number $\left(\frac{1}{3} - 5p\right)$ is irrational.

- i. State the negation of the statement above. **1**
- ii. Hence prove the given statement by contradiction. **2**

- (b) Rosa's office chair oscillates along a straight path with an amplitude of 3 cm.

The acceleration $\ddot{x}$ cm/s² of the chair is

$$\ddot{x} = 20 - 4x$$

where x is the displacement of the chair from her desk.

- i. Briefly explain why the chair is undergoing simple harmonic motion. **1**
- ii. By integrating $\ddot{x}$ with respect to x , show that the velocity of the chair is given by **2**

$$v^2 = -4\left((x - 5)^2 - 9\right)$$

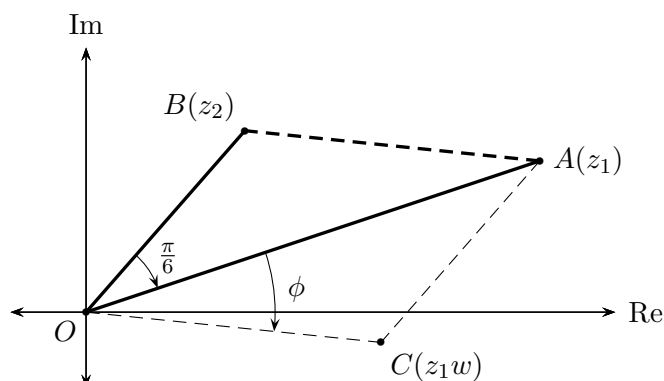
- iii. Find the fastest speed that the chair reaches. **1**

Given that the chair starts at the position closest to the desk:

- iv. Determine a possible equation for the displacement of the chair. **1**
- v. Determine whether the chair is moving towards or away from the desk after $\frac{37\pi}{24}$ seconds. Justify your answer. **2**

Examination continues overleaf...

- (c) The complex numbers z_1 and z_2 are represented by the points A and B respectively. In $\triangle OAB$, $OA = 2OB$ and $\angle AOB = \frac{\pi}{6}$ as shown in the Argand diagram.



It is given that $z_1 = r(\cos \theta + i \sin \theta)$, where $r \in \mathbb{R}$ and $0 < \theta < \frac{\pi}{2}$.

- i. Express z_2 in exponential form, in terms of r and θ . **2**

Let C be the point such that $ABOC$ is a parallelogram and $\angle AOC = \phi$, where $0 < \phi < \frac{\pi}{2}$. Consider the complex number w for which $z_1 w$ represents C .

- ii. By finding w , calculate ϕ correct to two decimal places. **3**

Question 13 (14 marks)**Marks**

- (a) Consider the region defined by

$$|z - 2 + 2i| \leq 4 \quad \text{and} \quad -\frac{\pi}{3} \leq \text{Arg}(z - 1 + 3i) \leq \frac{\pi}{4}$$

- i. On an Argand diagram, clearly sketch the region above. **3**

Let w represent any complex number that lies in the given region.

- ii. Find the possible range of values for $\text{Arg}(w)$, giving your answer correct to two decimal places. **3**

- (b) Prove that $|a - b| + |c - b| + |c - d| \geq a - d$. **2**

- (c) By considering the contrapositive, prove that for all $a, b \in \mathbb{Z}^+$ and where $n \in \mathbb{Z}$,
If $a + b \geq 2n + 1$, then $a \geq n + 1$ or $b \geq n + 1$. **2**

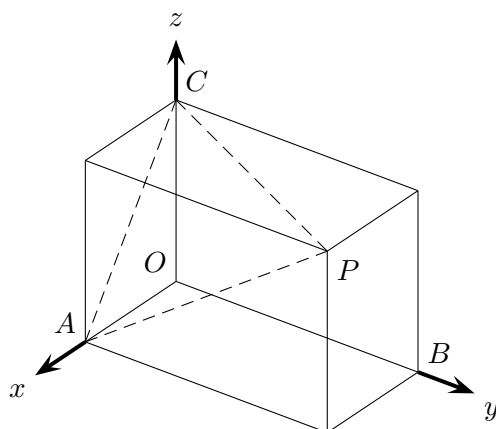
- (d) Use mathematical induction to prove that, for all integers $n \geq 3$, **4**

$$(3n)! < (n!)^3 \times 27^n$$

Examination continues overleaf...

Question 14 (14 marks)**Marks**

- (a) Consider the rectangular prism in the diagram.

3

Let $\underline{a}$, $\underline{b}$ and $\underline{c}$ be the non-zero position vectors of the points A , B and C .

Using vector methods, prove that triangle ACP **cannot** be a right-angled triangle.

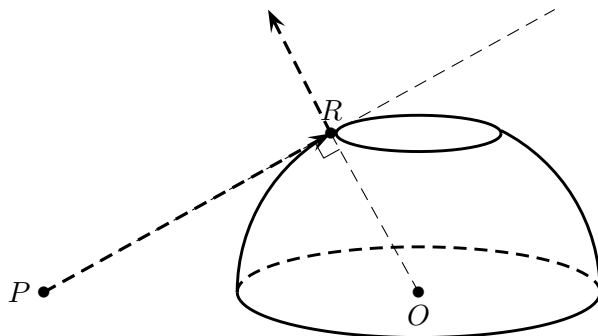
- (b) Find the exact value of

4

$$\int_{3\sqrt{2}}^6 \frac{dx}{x^3 \sqrt{x^2 - 9}}$$

- (c) A rock band is performing at a small arena which is shaped as a hemi-sphere and has a diameter of 26 metres. Let the centre of the arena be the origin O .

A mirror is placed at point $R(a, b, c)$ on the circular opening of the arena's roof, which is h metres above the ground. As a part of the lighting effects, a light will be shone from point $P(-46, 5, 0)$ outside the arena, to the mirror at R .



- i. Given that $PR \perp OR$, prove that $46a - 5b + 169 = 0$. 2
- ii. If the mirror at R is directly above a seat in the arena located at $(k, -3, 0)$, find the coordinates of R . 2

The mirror is angled so that the light is reflected upwards in the direction of $\overrightarrow{OR}$. Fireworks are shot along the path

$$\vec{f} = \begin{pmatrix} -14 \\ -10 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}, \quad \mu \in \mathbb{R}$$

- iii. Determine whether the trajectory of the fireworks will cross through the reflected ray of light. 3

Examination continues overleaf...

Question 15 (16 marks)**Marks**(a) Let a , b and c be positive real numbers.i. Prove that $a^2 + b^2 + c^2 \geq ab + ac + bc$. **2**ii. Hence or otherwise, prove that $a^3 + b^3 + c^3 \geq 3abc$. **1**iii. Hence prove that $ac(a + c) + ab(a + b) + bc(b + c) \geq 6abc$. **2**

(b) Consider the following series:

$$C = 1 + \cos \theta + \cos 2\theta + \cdots + \cos(n-1)\theta$$

$$S = \sin \theta + \sin 2\theta + \cdots + \sin(n-1)\theta$$

Given that $z = \cos \theta + i \sin \theta$:i. Show that $C + iS = \frac{1 - z^n}{1 - z}$ where $z \neq 1$. **2**ii. Hence show that **2**

$$\sin \theta + \sin 2\theta + \cdots + \sin(n-1)\theta = \frac{(\sin \theta - \sin n\theta) + \sin [(n-1)\theta]}{2(1 - \cos \theta)}$$

(c) Let $I_{3n+2} = \int_0^1 x^{3n+2} e^{x^3} dx$, where $n \in \mathbb{Z}$ and $n \geq 0$.i. Prove that $I_{3n+2} = \frac{e}{3} - nI_{3n-1}$. **2**ii. Hence find I_8 as an exact value. **2**iii. Using part (i), prove that **3**

$$3 \int_0^1 (1 + x^3) x^{3n-1} e^{x^3} dx \leq e$$

Question 16 (14 marks)**Marks**

- (a) i. By considering the a possible graph of $y = f(x)$, prove that: **3**

If f is a continuous function where $f'(x) > 0$, $f''(x) < 0$ and $f(0) = 0$, then

$$\int_0^a f(x) dx + \int_0^b f^{-1}(y) dy \geq ab$$

where f^{-1} is the inverse function of f , $a \geq 0$ and $b \geq 0$.

- ii. Hence prove that

$$\tan^{-1}\left(\frac{1}{2}\right) \geq \frac{\pi}{6} + \ln\left(\frac{15}{16}\right)$$

4

- (b) A object with a mass of 2 kg is at a height of 90 metres above the ground. It is dropped from rest and falls vertically downwards through a gaseous medium which exerts a resistance to the motion equal to $\frac{v^2}{45}$ newtons, where $v \text{ ms}^{-1}$ is the velocity of the particle.

After falling 25 metres, the object exits the gaseous medium and continues to fall vertically downwards, subject to air resistance equivalent to $\frac{v}{10}$ newtons.

The acceleration due to gravity is 10 ms^{-2} .

- i. Show that object exits the gaseous medium approximately 2.34 seconds after it was dropped. **4**

At the same time that the first object was dropped, a identical second object is launched from the ground directly below. The second object is launched vertically upwards at a speed of 40 ms^{-1} such that it will collide with the first object.

- ii. Determine in which medium would the two object collide. Justify your answer with the appropriate calculations. **3**

End of paper.

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Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

NESA STUDENT #:

Class (please ✓)

☐ 12MXX.1 – Ms Ham

☐ 12MXX.2 – Mr Ho

Directions for multiple choice answers

- Read each question and its suggested answers.
- Select the alternative (A), (B), (C), or (D) that best answers the question.
- Mark only one circle per question. There is only *one* correct choice per question.
- Fill in the response circle completely, using blue or black pen, e.g.

(A) (B) ● (D)

- If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) (B) ~~●~~ ●

- If you continue to change your mind, write the word **correct** and clearly indicate your final choice with an arrow as shown below:

(A) (B) ~~●~~ ~~●~~ ^{correct}

1 – (A) (B) (C) (D)

2 – (A) (B) (C) (D)

3 – (A) (B) (C) (D)

4 – (A) (B) (C) (D)

5 – (A) (B) (C) (D)

6 – (A) (B) (C) (D)

7 – (A) (B) (C) (D)

8 – (A) (B) (C) (D)

9 – (A) (B) (C) (D)

10 – (A) (B) (C) (D)

Sample Band E4 Responses

1. (D) 2. (B) 3. (C) 4. (B) 5. (D)
6. (A) 7. (C) 8. (D) 9. (D) 10. (B)

Note: Working out is not marked for MCQ, though solutions are provided as a learning opportunity on mathematical writing.

Question 1

If P , Q and R are collinear, $2\overrightarrow{PQ} = \overrightarrow{QR}$:

$$-8 = \lambda + 2$$

$$16 = \mu - 5$$

$$\therefore \lambda = -10 \quad \text{and} \quad \mu = 21$$

Correct option: (D)

Question 2

Using the direction vectors of lines r_1 and r_2 respectively:

$$\cos \theta = \frac{\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -4 \\ -2 \end{pmatrix}}{\sqrt{(-1)^2 + 2^2 + 1^2} \sqrt{3^2 + (-4)^2 + (-2)^2}}$$

$$\therefore \theta = \cos^{-1} \left(\frac{-13}{\sqrt{174}} \right)$$

Correct option: (B)

Question 3

- Six unique sixth roots of unity, evenly spaced apart by $\frac{\pi}{3}$ $\rightarrow$ not (B) or (D)
- $e^{i0} = 1$ is a root of $z^6 = 1$
- $e^{-i\pi} = e^{i\pi} = -1$ for $k = -3$ and $k = 3$ $\rightarrow$ not (A)

Correct option: (C)

Question 4

Negating each component:

$$\begin{array}{lll} \forall x \in \mathbb{R} & \rightarrow & \exists x \in \mathbb{R} \\ \exists y \in \mathbb{R} & \rightarrow & \forall y \in \mathbb{R} \\ x^y > 1 & \rightarrow & x^y \leq 1 \end{array}$$

Negation: $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}$ such that $x^y \leq 1$

Correct option: (B)

Question 5

Testing option (B) and (D) and splitting the fraction:

$$\begin{aligned} \frac{d}{dx} \left(\frac{1}{2} v^2 \right) &= \frac{d}{dx} \left(-\frac{6}{x^2} + \frac{4}{x} + C \right) \\ &= \frac{12}{x^3} - \frac{4}{x^2} \\ a &= \frac{12 - 4x}{x^3} \end{aligned}$$

$$\begin{aligned} \frac{1}{2} v^2 &= -\frac{6}{x^2} + \frac{4}{x} + C \\ v^2 &= -\frac{12}{x^2} + \frac{8}{x} + C \end{aligned}$$

When $x = 6, v = 0$:

$$\begin{aligned} v^2 &= -\frac{12}{6^2} + \frac{8}{6} + C \\ 0 &= 1 + C \\ C &= 0 \end{aligned}$$

$$\therefore v^2 = -\frac{12}{6^2} + \frac{8}{6} - 1$$

Correct option: (D)

Question 6

Given $\omega^3 = 1$:

$$\begin{aligned}(\omega - 1)(\omega^2 + \omega + 1) &= 0 \\ \omega^2 + \omega + 1 &= 0 && \text{(given } \operatorname{Im}(\omega) \neq 0) \\ \omega + 1 &= -\omega^2\end{aligned}$$

$$\begin{aligned}(1 + 2\omega - \omega^2 + \omega^3)^5 &= (1 + 2\omega - \omega^2 + 1)^5 \\ &= (-2\omega^2 - \omega^2)^5 \\ &= -243\omega^{10} \\ &= -243(\omega \cdot (\omega^3)^2) \\ &= -243\omega\end{aligned}$$

Correct option: (A)

Question 7

$$\begin{aligned}\frac{8\bar{z}z}{(\bar{z} - z)^2} &= \frac{8|z|^2}{[-2\operatorname{Im}(z)i]^2} \\ &= \frac{8([\operatorname{Re}(z)]^2 + [\operatorname{Im}(z)]^2)}{-4[\operatorname{Im}(z)]^2} \\ &= -2\left(\frac{[\operatorname{Re}(z)]^2}{[\operatorname{Im}(z)]^2} + 1\right) \\ &= -2 - 2\left(\frac{\operatorname{Re}(z)}{\operatorname{Im}(z)}\right)^2\end{aligned}$$

Correct option: (C)

Question 8

In the given graph, as $z \rightarrow \infty$, $x \rightarrow 0^+$ and $y \rightarrow 0^+$. By assuming $z \rightarrow \infty$ in the given options:

- Option (A): $t \rightarrow -\infty$, but then $x \rightarrow -\infty$.
- Option (B): $t \rightarrow \infty$, but then $x \rightarrow \infty$.
- Option (C): $t \rightarrow \infty$, but then $y \rightarrow 0^-$.
- Option (D): $t \rightarrow -\infty$, and $x \rightarrow 0^+$ and $y \rightarrow 0^+$.

Correct option: (D)

Question 9

Using the property $\int_0^a f(x) dx = \int_0^a f(a-x) dx$:

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \frac{x}{1 + \sin x + \cos x} dx &= \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2} - x}{1 + \sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2} - x}{1 + \cos x + \sin x} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}}{1 + \sin x + \cos x} dx - \int_0^{\frac{\pi}{2}} \frac{x}{1 + \sin x + \cos x} dx \end{aligned}$$

Correct option: (D)

Question 10

Let:

- p be 'Period 0 is cancelled'
- q be 'There is an excursion'

Rewriting the statement: $(p \Rightarrow q) \wedge (q \Rightarrow p)$

Negation of the statement: $(p \wedge q) \vee (q \wedge p)$

Correct option: (B)

Question 11 ()

(a) (2 marks)

- ✓ [1] for an equivalent fraction using the conjugate of the denominator
 ✓ [1] for the correct answer

$$\begin{aligned}\frac{-3+i}{1-2i} &= \frac{(-3+i)(1+2i)}{(1-2i)(1+2i)} && \checkmark \\ &= \frac{-5-5i}{1+4} \\ &= -1-i && \checkmark\end{aligned}$$

(b) (2 marks)

- ✓ [1] for stating $\text{Arg}(z_2) + \text{Arg}(z_3) = \text{Arg}(z_2 z_3)$
 ✓ [1] for the correct answer

$$\begin{aligned}\text{Arg}(z_2) + \text{Arg}(z_3) &= \text{Arg}(z_2 z_3) && \checkmark \\ \pi - \tan^{-1}\left(\frac{2\sqrt{3}}{2}\right) + \text{Arg}(z_3) &= \frac{\pi}{3} \\ \frac{2\pi}{3} + \text{Arg}(z_3) &= \frac{\pi}{3} \\ \therefore \text{Arg}(z_3) &= -\frac{\pi}{3} && \checkmark\end{aligned}$$

(c) (2 marks)

- ✓ [1] for one correct root
 ✓ [1] for all correct roots with sufficient working out

$2+3i$ is also a root since $p(z)$ real coefficients. $\checkmark$

By sum of roots:

$$\begin{aligned}(2-3i) + (2+3i) + z_3 &= -(-8) \\ z_3 &= 4 && \checkmark\end{aligned}$$

$\therefore$ Roots z_2 and z_3 are $2-3i$ and $z_3 = 4$.

(d) (2 marks)

- ✓ [1] for correctly equating the real and imaginary components
- ✓ [1] for all correct answers

Let $(x + iy)^2 = 21 - 20i$.

$$x^2 - y^2 + (2xy)i = 21 - 20i$$

Equating real and imaginary components:

$$x^2 - y^2 = 21$$

$$2xy = -20$$

$$xy = -10 \quad \checkmark$$

By inspection: $x = \pm 5$ and $y = \mp 2$ $\therefore$ The square roots are $5 - 2i$ and $-5 + 2i$. $\checkmark$

(e) (2 marks)

- ✓ [1] for correctly substituting an expression for k as an odd integer
- ✓ [1] for correctly proving the result with sufficient justifications

Let $k = 2a + 1$, where $a \in \mathbb{Z}$.

$$\begin{aligned}
 \frac{3}{2}k^2 - 6k - \frac{63}{2} &= \frac{3}{2}\left((2a+1)^2 - 4(2a+1) - 21\right) \quad \checkmark \\
 &= \frac{3}{2}(4a^2 - 4a - 24) \\
 &= 6(a^2 - a - 6) \\
 &= 6(a(a-1) - 6) \\
 &= 6(2m - 6) \quad \text{where } m \in \mathbb{Z} \\
 &\quad \text{(since one of the consecutive integers } a-1 \text{ or } a \text{ must be negative)} \\
 &= 12(m-3) \quad \text{where } m-3 \in \mathbb{Z}
 \end{aligned}$$

$\therefore \frac{3}{2}k^2 - 6k - \frac{63}{2}$ is divisible by 12 for any positive odd integer k . $\checkmark$

(f) i. (2 marks)

✓ [1] for significant progress

✓ [1] for correctly showing the required result

$$\frac{9x}{(x^2+2)(x+1)} = \frac{Ax+B}{x^2+2} + \frac{C}{x+1}$$

$$9x = (Ax+B)(x+1) + C(x^2+2)$$

$$\text{Let } x = -1: \quad -9 = 3C \quad \rightarrow \quad C = -3 \quad \checkmark$$

$$\text{Let } x = 0: \quad 0 = B - 6 \quad \rightarrow \quad B = 6$$

$$\text{Let } x = 1: \quad 9 = 2(A+6) - 9 \quad \rightarrow \quad A = 3$$

$$\therefore \frac{9x}{(x^2+2)(x+1)} = \frac{3x+6}{x^2+2} - \frac{3}{x+1}$$

ii. (2 marks)

✓ [2] for the correct integral

$$\begin{aligned} \int \frac{9x}{(x^2+2)(x+1)} dx &= \int \frac{3x+6}{x^2+2} - \frac{3}{x+1} dx \\ &= \int \frac{3x}{x^2+2} + \frac{6}{x^2+2} - \frac{3}{x+1} dx \\ &= \frac{3}{2} \ln|x^2+2| + \frac{6}{\sqrt{2}} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right) - 3 \ln|x+1| + C \quad \checkmark\checkmark \end{aligned}$$

(g) (3 marks)

✓ [1] for identifying a correct substitution

✓ [1] for correctly substituting the x variable for the u variable

✓ [1] for the correct answer

$$\text{Let } u = 1 + \sqrt{2x}. \quad \checkmark$$

$$\frac{du}{dx} = \frac{1}{2} \cdot \frac{2}{\sqrt{2x}}$$

$$\sqrt{2x} du = dx$$

$$\begin{aligned} \int \frac{2}{1+\sqrt{2x}} dx &= 2 \int \frac{1}{u} \cdot \sqrt{2x} du \\ &= 2 \int \frac{u-1}{u} du \\ &= 2 \int 1 - \frac{1}{u} du \quad \checkmark \\ &= 2u - 2 \ln|u| + C \\ &= 2 + 2\sqrt{2x} - 2 \ln(1 + \sqrt{2x}) + C \quad \checkmark \end{aligned}$$

Question 12 ()

(a) i. (1 mark)

✓ [1] for the correct negation

There exists an irrational number p such that $\frac{1}{3} - 5p$ is rational. ✓

ii. (2 marks)

✓ [1] for correctly expressing p as a rational number

✓ [2] for correctly explaining the contradiction and thus proving by contradiction.

Assume the negation from part (i) is true, i.e. $p \notin \mathbb{Q}$ and $\frac{1}{3} - 5p \in \mathbb{Q}$.

Let $a, b \in \mathbb{Z}$, $b \neq 0$ and HCF of a and b be is 1, such that

$$\begin{aligned}\frac{1}{3} - 5p &= \frac{a}{b} \\ -5p &= \frac{3a - b}{3b} \\ p &= \frac{b - 3a}{15b} \quad \checkmark\end{aligned}$$

So p is always rational since $b - 3a \in \mathbb{Z}$ and $15b \in \mathbb{Z}$, given that $a, b \in \mathbb{Z}$ (and $b \neq 0$).

But $p \notin \mathbb{Q}$ from assumption.

Hence by contradiction, the given statement is true. ✓

(b) i. (1 mark)

✓ [1] for correctly showing the acceleration can be expressed as $\ddot{x} = -n^2(x - c)$

$$\begin{aligned}\ddot{x} &= 20 - 4x \\ &= -2^2(x - 5) \quad \checkmark\end{aligned}$$

$\therefore$ The chair in SHM as its acceleration can be expressed in the form $\ddot{x} = -n^2(x - c)$ with a centre of motion $c = 5$ and frequency $n = 2$.

ii. (2 marks)

✓ [1] for the correct indefinite integral of $\frac{1}{2}v^2$ in terms of x

✓ [2] for the correct final answer

At extremities $x = 5 \pm 3$, $v = 0$.

$$\begin{aligned}\frac{d}{dx} \left(\frac{1}{2}v^2 \right) &= 20 - 4x \\ \int_0^v d \left(\frac{1}{2}v^2 \right) &= \int_2^x 20 - 4x \, dx \\ \left[\frac{1}{2}v^2 \right]_0^v &= \left[20x - 2x^2 \right]_2^x \quad \checkmark \\ \frac{1}{2}v^2 &= 20x - 2x^2 - 32 \\ v^2 &= 40x - 4x^2 - 64 \\ &= -4(x^2 - 10x + 16) \\ &= -4(x^2 - 10x + 25 - 9) \\ \therefore v^2 &= -4((x - 5)^2 - 9) \quad \checkmark\end{aligned}$$

iii. (1 mark)

✓ [1] for the correct answer

Fastest speed occurs at $x = 5$.

$$\begin{aligned}v^2 &= 36 & (\text{from ii.}) \\ |v| &= 6\end{aligned}$$

$\therefore$ The fastest speed reached is 6 metres.

iv. (1 mark)

✓ [1] for a correct displacement equation

Given $a = 3$, $n = 2$, $c = 5$ and $x(0) = 2$.

$$x = -3 \cos(2t) + 5 \quad \checkmark$$

v. (2 marks)

- ✓ [1] for calculating the velocity at $t = \frac{37\pi}{24}$, or describing the motion using $T = \pi$
- ✓ [1] for the correct conclusion from relevant working out

$$\dot{x} = 6 \sin(2t)$$

When $t = \frac{37\pi}{24}$,

$$\dot{x} = -1.55 \dots \quad \checkmark$$

$\therefore$ The particle is moving towards the desk. $\checkmark$

Alternate method:

The motion has a period of π seconds, and $\frac{3\pi}{2} < \frac{37\pi}{24} < 2\pi$.

The chair starts at the minimum displacement $x = 2$.

After exactly $\frac{3\pi}{2}$ seconds, it will be at the maximum displacement $x = 8$. $\checkmark$

Hence the chair will be moving towards the desk at $t = \frac{37\pi}{24}$ seconds. $\checkmark$

(c) i. (2 marks)

- ✓ [1] for multiplying to a correct modulus OR argument
- ✓ [1] for the correct answer in exponential form

$$z_1 = re^{i\theta}$$

$$z_2 = re^{i\theta} \times \frac{1}{2}e^{i\frac{\pi}{6}} \quad \checkmark$$

$$\therefore z_2 = \frac{1}{2}re^{i(\theta+\frac{\pi}{6})} \quad \checkmark$$

ii. (3 marks)

- ✓ [1] for a correct expression for $z_1 w$
- ✓ [1] for finding w
- ✓ [1] for the correct answer

Opposite sides of a parallelogram are equal and parallel:

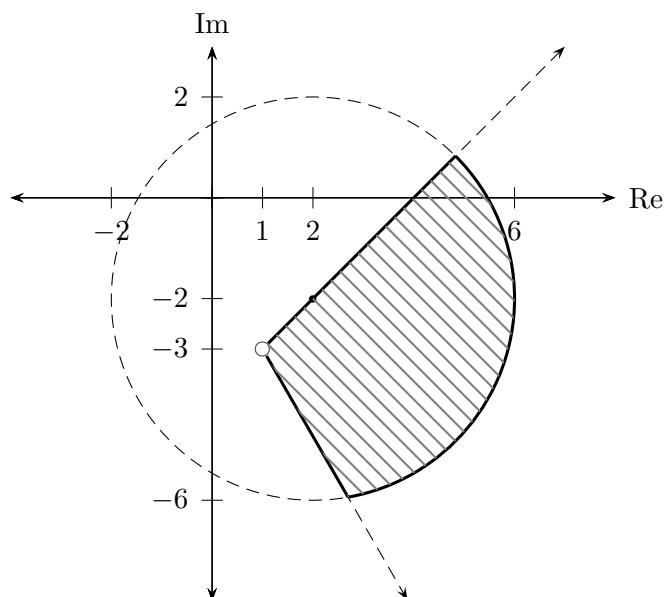
$$\overrightarrow{BA} = \overrightarrow{OC}$$

$$\begin{aligned} z_1 w &= z_1 - z_2 \quad \checkmark \\ &= re^{i\theta} - \frac{1}{2}re^{i(\theta+\frac{\pi}{6})} \\ re^{i\theta}w &= re^{i\theta}\left(1 - \frac{1}{2}e^{i\frac{\pi}{6}}\right) \\ w &= 1 - \frac{1}{2}e^{i\frac{\pi}{6}} \quad \checkmark \\ &= 1 - \frac{1}{2}\cos\frac{\pi}{6} - \frac{1}{2}i\sin\frac{\pi}{6} \\ &= \left(1 - \frac{\sqrt{3}}{4}\right) - \frac{1}{4}i \\ \arg(w) &= -0.42 \text{ radians} \\ \therefore \phi &= 0.42 \text{ radians} \quad \checkmark \end{aligned}$$

Question 13 ()

(a) i. (3 marks)

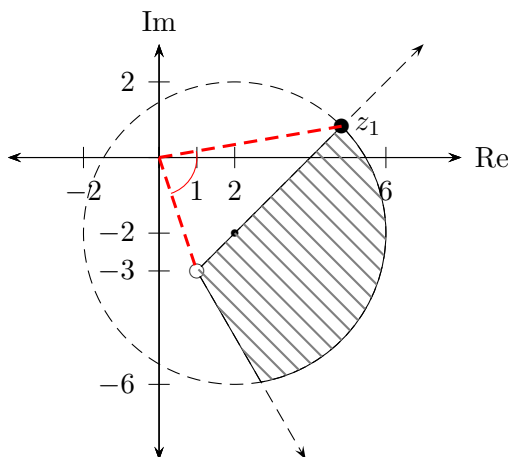
- ✓ [1] for the correct circle representing $|z - 2 + 2i| \leq 4$
- ✓ [1] for the correct rays representing $-\frac{\pi}{3} \leq \arg(z - 1 + 3i) \leq \frac{\pi}{4}$ with $(1 - 3i)$ excluded
- ✓ [1] for the correct shaded region



ii. (3 marks)

- ✓ [1] for identifying the values of w for which $\text{Arg}(w)$ is a minimum and maximum
- ✓ [1] for the correct minimum or maximum value of $\text{Arg}(w)$
- ✓ [1] for the correct range of values for $\text{Arg}(w)$ with an exclusive lower limit

Let z_1 be a complex number such that $|z_1 - 2 + 2i| = 4$ and $\text{Arg}(z_1 - 1 + 3i) = \frac{\pi}{4}$.



For z_1 :

$$(x - 2)^2 + (y + 2)^2 = 16 \quad (1)$$

$$y = x - 4 \quad (2)$$

Substitute (2) into (1):

$$(x - 2)^2 + (x - 2)^2 = 16$$

$$(x - 2)^2 = 8$$

$$x = 2 + 2\sqrt{2} \quad (\text{since } \text{Re}(z_1) > 0)$$

Substitute $x = 2 + 2\sqrt{2}$ into (3):

$$y = -2 + 2\sqrt{2}$$

$$z_1 = (2 + 2\sqrt{2}) + i(-2 + 2\sqrt{2})$$

$$\text{Arg}(z_1) = 0.17 \text{ radians} \quad \checkmark$$

From the Argand diagram:

$$\text{Arg}(1 - 3i) < \text{Arg}(w) \leq \text{Arg}(z_1) \quad \checkmark$$

$$\therefore -1.25 < \text{Arg}(w) \leq 0.17 \quad \checkmark$$

(b) (2 marks)

- ✓ [1] for correctly applying the triangle inequality once
- ✓ [1] for correctly proving the required result with justification for $|a - d| \geq a - d$

$$\mathbf{RTP:} \quad |a - b| + |c - b| + |c - d| \geq a - d$$

$$\begin{aligned} |a - b| + |c - b| + |c - d| &= (|a - b| + |b - c|) + |c - d| \\ &\geq |a - b + b - c| + |c - d| && \text{(triangle inequality)} \\ &= |a - c| + |c - d| \\ &\geq |a - d| && \text{(triangle inequality)} \end{aligned}$$

$|a - d| \geq a - d$ since:

- if $a \geq d$, then $|a - d| = a - d$
- if $a < d$, then $|a - d| > a - d$

$$\therefore |a - b| + |c - b| + |c - d| \geq a - d$$

(c) (2 marks)

- ✓ [1] for the correct contrapositive
- ✓ [1] for correctly proving the required result with sufficient conclusions

Contrapositive: If $a < n + 1$ and $b < n + 1$, then $a + b < 2n + 1$

for all $a, b \in \mathbb{Z}$ and where $n \in \mathbb{Z}$

Since $a, b \in \mathbb{Z}$, $a \leq n$ and $b \leq n$.

$$\begin{aligned} a + b &\leq 2n \\ &< 2n + 1 \end{aligned}$$

$\therefore$ The contrapositive is true.

Hence for all $a, b \in \mathbb{Z}$ and where $n \in \mathbb{Z}$, if $a + b \geq 2n + 1$, then $a \geq n + 1$ or $b \geq n + 1$.

(d) (3 marks)

- ✓ [1] for finding r_{AM} , or equivalent merit.
- ✓ [2] for substantial progress in setting up a system of equations by equating r_{AM} with r_{BT} .
- ✓ [3] for fully correct solution and justification.

Let $P(n)$ be the proposition.

Base case: For $n = 3$,

$$\begin{aligned}\text{LHS} &= 9! \\ &= 362\,880\end{aligned}$$

$$\begin{aligned}\text{RHS} &= (3!)^3 \times 27^3 \\ &= 4\,251\,528\end{aligned}$$

$\therefore$ LHS < RHS, and $P(3)$ is true.

Inductive step: Assume $P(k)$ is true.

$$(3k)! < (k!)^3 \times 27^k$$

Consider $n = k + 1$:

$$\mathbf{RTP:} \quad (3k+3)! < ((k+1)!)^3 \times 27^{k+1}$$

$$\begin{aligned}\text{LHS} &= (3k+3)(3k+2)(3k+1) \cdot (3k)! \\ &< (3k+3)(3k+2)(3k+1) \left((k!)^3 \times 27^k \right) && \text{(from assumption)} \\ &= 3^3(k+1) \left(k + \frac{2}{3} \right) \left(k + \frac{1}{3} \right) \cdot (k!)^3 \times 27^k \\ &= (k+1) \left(k + \frac{2}{3} \right) \left(k + \frac{1}{3} \right) \cdot (k!)^3 \times 27^{k+1} \\ &< (k+1)^3 (k!)^3 \times 27^{k+1} && \text{(since } k \geq 3\text{)} \\ &= ((k+1)!)^3 \times 27^{k+1}\end{aligned}$$

$\therefore$ LHS < RHS, and $P(k+1)$ is true if $P(k)$ is true.

Hence by mathematical induction, $P(n)$ is true for $n \geq 3$.

Question 14 ()

(a) (3 marks)

- ✓ [1] for correctly showing that one pair of sides cannot be perpendicular OR significant progress in showing no interiors can be right-angles
- ✓ [1] for correctly showing that two pairs of sides cannot be perpendicular
- ✓ [1] for correctly proving the required result

Let $\overrightarrow{OA} = \begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} 0 \\ b \\ 0 \end{pmatrix}$ and $\overrightarrow{OC} = \begin{pmatrix} 0 \\ 0 \\ c \end{pmatrix}$ where $a, b, c \in \mathbb{Z}^+$.

$$\begin{aligned} \overrightarrow{AC} \cdot \overrightarrow{AP} &= \begin{pmatrix} -a \\ 0 \\ c \end{pmatrix} \cdot \begin{pmatrix} 0 \\ b \\ c \end{pmatrix} \\ &= c^2 \end{aligned}$$

$\therefore AC$ cannot be perpendicular to AP since c is non-zero. ✓

$$\begin{aligned} \overrightarrow{CA} \cdot \overrightarrow{CP} &= \begin{pmatrix} a \\ 0 \\ -c \end{pmatrix} \cdot \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} \\ &= a^2 \end{aligned}$$

$\therefore CA$ cannot be perpendicular to CP since a is non-zero. ✓

$$\begin{aligned} \overrightarrow{PA} \cdot \overrightarrow{PC} &= \begin{pmatrix} 0 \\ -b \\ -c \end{pmatrix} \cdot \begin{pmatrix} -a \\ -b \\ 0 \end{pmatrix} \\ &= b^2 \end{aligned}$$

$\therefore PA$ cannot be perpendicular to PC since b is non-zero.

Hence triangle ACP cannot be a right-angled triangle. ✓

Alternate method:

Let $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$ and $\overrightarrow{OC} = \underline{c}$.

Since coordinate axes are perpendicular to each other,

$$\underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{c} = \underline{a} \cdot \underline{c} = 0$$

$$\begin{aligned}\overrightarrow{AC} \cdot \overrightarrow{AP} &= (\underline{c} - \underline{a}) \cdot (\underline{b} + \underline{c}) \\ &= \underline{b} \cdot \underline{c} + |\underline{c}|^2 - \underline{a} \cdot \underline{b} - \underline{a} \cdot \underline{c} \\ &= |\underline{c}|^2\end{aligned}$$

$\therefore AC$ cannot be perpendicular to AP since $|\underline{c}|$ is non-zero. $\checkmark$

$$\begin{aligned}\overrightarrow{CA} \cdot \overrightarrow{CP} &= (\underline{a} - \underline{c}) \cdot (\underline{a} + \underline{b}) \\ &= |\underline{a}|^2 + \underline{a} \cdot \underline{b} - \underline{a} \cdot \underline{c} - \underline{b} \cdot \underline{c} \\ &= |\underline{a}|^2\end{aligned}$$

$\therefore CA$ cannot be perpendicular to CP since $|\underline{a}|$ is non-zero. $\checkmark$

$$\begin{aligned}\overrightarrow{PA} \cdot \overrightarrow{PC} &= (-\underline{b} - \underline{c}) \cdot (-\underline{a} - \underline{b}) \\ &= \underline{a} \cdot \underline{b} + |\underline{b}|^2 + \underline{a} \cdot \underline{c} + \underline{b} \cdot \underline{c} \\ &= |\underline{b}|^2\end{aligned}$$

$\therefore PA$ cannot be perpendicular to PC since $|\underline{b}|$ is non-zero.

Hence triangle ACP cannot be a right-angled triangle. $\checkmark$

(b) (4 marks)

- ✓ [1] for selecting a correct substitution with significant progress
- ✓ [1] for correctly simplifying the expression after substitution
- ✓ [1] for the correct integral from the double angle formula of $\cos^2 \theta$
- ✓ [1] for the correct final answer

Let $x = 3 \sec \theta$.

$$\begin{aligned}\frac{dx}{d\theta} &= \frac{-3(-\sin \theta)}{\cos^2 \theta} \\ dx &= 3 \sec \theta \tan \theta d\theta \quad \checkmark\end{aligned}$$

$$\begin{aligned}x = 6 &\quad \rightarrow \quad \theta = \frac{\pi}{3} \\ x = 3\sqrt{2} &\quad \rightarrow \quad \theta = \frac{\pi}{4}\end{aligned}$$

$$\begin{aligned}\int_{3\sqrt{2}}^6 \frac{dx}{x^3 \sqrt{x^2 - 9}} &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{3 \sec \theta \tan \theta}{27 \sec^3 \theta \sqrt{9 \sec^2 \theta - 9}} d\theta \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan \theta}{9 \sec^2 \theta (3 \tan \theta)} d\theta \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{27 \sec^2 \theta} d\theta \quad \checkmark \\ &= \frac{1}{27} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \cos^2 \theta d\theta \\ &= \frac{1}{27} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{2} (\cos 2\theta + 1) d\theta \\ &= \frac{1}{54} \left[\frac{1}{2} \sin 2\theta + \theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} \quad \checkmark \\ &= \frac{1}{54} \left[\left(\frac{\sqrt{3}}{4} + \frac{\pi}{3} \right) - \left(\frac{1}{2} + \frac{\pi}{4} \right) \right] \\ &= \frac{1}{54} \left(\frac{\sqrt{3} - 2}{4} + \frac{\pi}{12} \right) \quad \checkmark \\ &= \frac{\pi + 3\sqrt{3} - 6}{648}\end{aligned}$$

(c) i. (2 marks)

✓ [1] for correctly simplifying $\overrightarrow{PR} \cdot \overrightarrow{OR}$

✓ [1] for correctly showing the required result by considering the sphere

$$\begin{aligned}\overrightarrow{PR} &= \begin{pmatrix} a \\ b \\ c \end{pmatrix} - \begin{pmatrix} -46 \\ 5 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} a+46 \\ b-5 \\ c \end{pmatrix}\end{aligned}$$

$$\overrightarrow{PR} \cdot \overrightarrow{OR} = 0 \quad (\text{given } PR \perp OR)$$

$$\begin{pmatrix} a+46 \\ b-5 \\ c \end{pmatrix} \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0$$

$$a^2 + 46a + b^2 - 5b + c^2 = 0 \quad \checkmark \quad (1)$$

Since R lies on the sphere:

$$\begin{aligned}x^2 + y^2 + z^2 &= 13^2 \\ a^2 + b^2 + c^2 &= 169\end{aligned} \quad (2)$$

Substitute (2) into (1):

$$\therefore 46a - 5b + 169 = 0 \quad \checkmark$$

ii. (2 marks)

✓ [1] for the correct x -coordinate for R ✓ [1] for the correct coordinates for R Let $b = -3$. From (i):

$$\begin{aligned}46a + 15 + 169 &= 0 \\ a &= -4 \quad \checkmark\end{aligned}$$

From (2):

$$\begin{aligned}(-4)^2 + (-3)^2 + c^2 &= 169 \\ c &= 12\end{aligned}$$

$$\therefore R(-4, -3, 12) \quad \checkmark$$

iii. (3 marks)

- ✓ [1] for correctly equating the vector equations of the two lines
- ✓ [1] for a correct set of values for λ and μ from the system of equations
- ✓ [1] for the correct conclusion with relevant working

Let $\underline{r}$ be the equation of line OR :

$$\underline{r} = \lambda \begin{pmatrix} -4 \\ -3 \\ 12 \end{pmatrix} \quad \text{where } \lambda \in \mathbb{R}$$

Let $\underline{f} = \underline{r}$:

$$\begin{pmatrix} -14 \\ -10 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} = \lambda \begin{pmatrix} -4 \\ -3 \\ 12 \end{pmatrix} \quad \checkmark$$

Equating the components:

$$-14 + 2\mu = -4\lambda \quad (3)$$

$$-10 + \mu = -3\lambda \quad (4)$$

$$5\mu = 12\lambda \quad (5)$$

(3) $- 2 \times$ (4) :

$$6 = 2\lambda$$

$$\lambda = 3$$

Substitute $\lambda = 3$ into (4):

$$\mu = 1 \quad \checkmark$$

Substitute $\lambda = 3$ and $\mu = 1$ into (5):

$$\text{LHS} = 5$$

$$\text{RHS} = 36$$

$$\text{LHS} \neq \text{RHS}$$

$\therefore$ No values of λ and μ such that $\underline{f} = \underline{r}$.

Hence the trajectory of the fireworks will **NOT** cross through the reflected ray of light. $\checkmark$

Question 15 ()

(a) i. (2 marks)

✓ [1] for correctly showing $a^2 + b^2 \geq 2ab$ or equivalent

✓ [1] for correctly showing the required result

$$\begin{aligned}
 (a - b)^2 &\geq 0 \\
 a^2 - 2ab + b^2 &\geq 0 \\
 a^2 + b^2 &\geq 2ab \quad \checkmark
 \end{aligned} \tag{1}$$

Similarly:

$$a^2 + c^2 \geq 2ac \tag{2}$$

$$b^2 + c^2 \geq 2bc \tag{3}$$

(1) + (2) + (3) :

$$\begin{aligned}
 2(a^2 + b^2 + c^2) &\geq 2(ab + ac + bc) \\
 \therefore a^2 + b^2 + c^2 &\geq ab + ac + bc \quad \checkmark
 \end{aligned} \tag{4}$$

ii. (3 marks)

✓ [1] for correctly showing the required result

Multiply (4) from (i), by $(a + b + c)$:

$$(a + b + c)(a^2 + b^2 + c^2) \geq (a + b + c)(ab + ac + bc)$$

$$a^3 + b^3 + c^3 + ab^2 + ac^2 + a^2b + bc^2 + a^2c + b^2c \geq a^2b + a^2c + ab^2 + b^2c + ac^2 + bc^2 + abc + abc + abc$$

$$\therefore a^3 + b^3 + c^3 \geq 3abc \quad \checkmark$$

iii. (2 marks)

✓ [1] for correctly multiplying two applications of part (ii)

✓ [1] for correctly showing the required result using (ii)

$$a^3 + b^3 + c^3 \geq 3abc \tag{5}$$

Let $a \rightarrow \frac{1}{a}$, $b \rightarrow \frac{1}{b}$ and $c \rightarrow \frac{1}{c}$:

$$\frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} \geq \frac{3}{abc} \tag{6}$$

Multiply (1) and (2):

$$\begin{aligned}
 (a^3 + b^3 + c^3) \left(\frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} \right) &\geq 9 \quad \checkmark \\
 1 + 1 + 1 + \frac{b^3 + c^3}{a^3} + \frac{a^3 + c^3}{b^3} + \frac{a^3 + b^3}{c^3} &\geq 9 \\
 b^3c^3(b^3 + c^3) + a^3c^3(a^3 + c^3) + a^3b^3(a^3 + b^3) &\geq 6a^3b^3c^3
 \end{aligned}$$

Let $a \rightarrow \sqrt[3]{a}$, $b \rightarrow \sqrt[3]{b}$ and $c \rightarrow \sqrt[3]{c}$:

$$\therefore bc(b + c) + ac(a + c) + ab(a + b) \geq 6abc \quad \checkmark$$

(b) i. (2 marks)

✓ [1] for correctly applying De Moivre's Theorem

✓ [1] for correctly showing the required result

$$\begin{aligned}
C + iS &= (1 + \cos \theta + \cos 2\theta + \cdots + \cos(n-1)\theta) + i(1 + \sin 2\theta + \cdots + \sin(n-1)\theta) \\
&= 1 + (\cos \theta + i \sin \theta) + (\cos 2\theta + i \sin 2\theta) + \cdots + (\cos(n-1)\theta + i \sin(n-1)\theta) \\
&= 1 + (\cos \theta + i \sin \theta) + (\cos \theta + i \sin \theta)^2 \cdots + (\cos \theta + i \sin \theta)^{n-1} \quad \checkmark \\
&= 1 + z + z^2 + \cdots + z^{n-1} \\
&= \frac{(1-z)(1+z+z^2+\cdots+z^{n-1})}{1-z}
\end{aligned}$$

$$\therefore C + iS = \frac{1-z^n}{1-z} \quad \checkmark$$

ii. (2 marks)

✓ [1] for correctly rationalising $C + iS$

✓ [1] for correctly equating the imaginary components to show the required result

$$\begin{aligned}
C + iS &= \frac{1-z^n}{1-z} \\
&= \frac{1-(\cos \theta + i \sin \theta)^n}{1-(\cos \theta + i \sin \theta)} \\
&= \frac{1-(\cos n\theta + i \sin n\theta)}{(1-\cos \theta) - i \sin \theta} \\
&= \frac{((1-\cos n\theta) - i \sin n\theta)((1-\cos \theta) + i \sin \theta)}{((1-\cos \theta) - i \sin \theta)((1-\cos \theta) + i \sin \theta)} \\
&= \frac{(1-\cos n\theta)(1-\cos \theta) + \sin n\theta \sin \theta + i(\sin \theta(1-\cos n\theta) - \sin n\theta(1-\cos \theta))}{(1-\cos \theta)^2 + \sin^2 \theta} \quad \checkmark
\end{aligned}$$

Equate imaginary components:

$$\begin{aligned}
S &= \frac{\sin \theta - \cos n\theta \sin \theta - \sin n\theta + \sin n\theta \cos \theta}{1 - 2\cos \theta + \cos^2 \theta + \sin^2 \theta} \\
&= \frac{(\sin \theta - \sin n\theta) + \sin(n\theta - \theta)}{2 - 2\cos \theta}
\end{aligned}$$

$$\therefore 1 + \sin 2\theta + \cdots + \sin(n-1)\theta = \frac{(\sin \theta - \sin n\theta) + \sin[(n-1)\theta]}{2(1-\cos \theta)} \quad \checkmark$$

(c) i. (2 marks)

✓ [1] for correctly applying integration by parts

✓ [1] for correctly proving the required result

$$\begin{aligned} \text{Let } u = x^{3n} &\rightarrow u' = 3nx^{3n-1} \\ v' = x^2 e^{x^3} &\rightarrow v = \frac{1}{3} e^{x^3} \end{aligned}$$

$$\begin{aligned} I_{3n+2} &= \int_0^1 x^{3n+2} e^{x^3} dx \\ &= \left[\frac{1}{3} x^{3n} e^{x^3} \right]_0^1 - \int_0^1 nx^{3n-1} e^{x^3} dx \quad \checkmark \\ &= \frac{1}{3} (e - 0) - nI_{3n-1} \\ \therefore I_{3n+2} &= \frac{e}{3} - nI_{3n-1} \quad \checkmark \end{aligned}$$

ii. (2 marks)

✓ [1] for correctly applying the reduction formula from part (i)

✓ [1] for the correct final answer

$$\begin{aligned} I_8 &= \frac{e}{3} - 2I_5 \quad \checkmark \\ &= \frac{e}{3} - 2 \left(\frac{e}{3} - I_2 \right) \\ &= -\frac{e}{3} + 2 \int_0^1 x^2 e^{x^3} dx \\ &= -\frac{e}{3} + 2 \left[\frac{1}{3} e^{x^3} \right]_0^1 \\ &= -\frac{e}{3} + \frac{2}{3} (e - 1) \\ \therefore I_8 &= \frac{e-2}{3} \quad \checkmark \end{aligned}$$

iii. (3 marks)

- ✓ [1] for correctly simplifying the given integral
- ✓ [1] for correctly showing that $I_{3n-1} \geq 0$
- ✓ [1] for correctly proving the required result with reference to $(n-1)$

$$\begin{aligned}
 3 \int_0^1 (1+x^3)x^{3n-1}e^{x^3} dx &= 3 \left(\int_0^1 x^{3n-1}e^{x^3} dx + \int_0^1 x^{3n+2}e^{x^3} dx \right) \\
 &= 3(I_{3n-1} + I_{3n+2}) \\
 &= 3 \left(I_{3n-1} + \frac{e}{3} - nI_{3n-1} \right) \quad (\text{from i.}) \\
 &= e - 3(n-1)I_{3n-1} \quad \checkmark
 \end{aligned}$$

For $0 \leq x \leq 1$:

$$\begin{aligned}
 x^{3n-1} &\geq 0 && (\text{since } n \geq 1) \\
 x^3 \geq 0 \quad \text{so} \quad e^{x^3} &> 0 && (\text{since } f(0) = 0) \\
 x^{3n-1}e^{x^3} &\geq 0
 \end{aligned}$$

$$\begin{aligned}
 \therefore \int_0^1 x^{3n-1}e^{x^3} dx &\geq 0 \\
 I_{3n-1} &\geq 0 \quad \checkmark \\
 3(n-1)I_{3n-1} &\geq 0 && (\text{since } n \geq 1)
 \end{aligned}$$

$$\text{Hence } 3 \int_0^1 (1+x^3)x^{3n-1}e^{x^3} dx \leq e \quad \checkmark$$

Question 16 ()

(a) i. (3 marks)

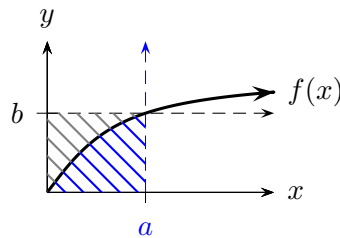
- ✓ [1] for correctly explaining one case with a graph
- ✓ [1] for correctly explaining two cases with graph(s)
- ✓ [1] for correctly proving the required result by explaining all cases

$\int_0^b f^{-1}(y) dy$ is the area bounded by $y = f(x)$, the y -axis, $y = 0$ and $y = b$:

$$f(x) = y \quad \rightarrow \quad x = f^{-1}(y)$$

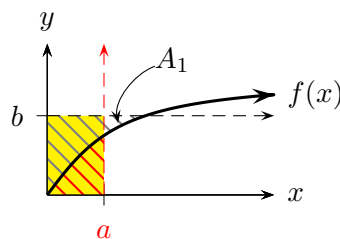
Consider the graph of a continuous function that is increasing and concave down:

Case 1: If $f(a) = b$,



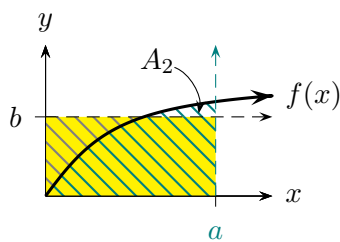
$$\int_0^a f(x) dx + \int_0^b f^{-1}(y) dy = ab \quad \checkmark$$

Case 2: If $f(a) < b$,



$$\int_0^a f(x) dx + \int_0^b f^{-1}(y) dy = ab + A_1 > ab \quad \checkmark$$

Case 3: If $f(a) > b$,



$$\int_0^a f(x) dx + \int_0^b f^{-1}(y) dy = ab + A_2$$

$$> ab$$

Hence if f is a continuous function where $f'(x) > 0$, $f''(x) < 0$ and $f(0) = 0$, then

$$\int_0^a f(x) dx + \int_0^b f^{-1}(y) dy \geq ab$$

where f^{-1} is the inverse function of f , $a \geq 0$ and $b \geq 0$. ✓

Trivial case: If $a = 0$ or $b = 0$, $ab = 0$.

$$\int_0^a f(x) dx \geq 0 \quad \text{or} \quad \int_0^b f^{-1}(y) dy \geq 0$$

$$\therefore \int_0^a f(x) dx + \int_0^b f^{-1}(y) dy \geq ab$$

ii. (4 marks)

- ✓ [1] for correctly identifying $f(x) = \tan^{-1} x$ as a suitable function
- ✓ [1] for the correct integration by parts
- ✓ [1] for a correct inequality for $\tan^{-1} a$
- ✓ [1] for correctly proving the required result

Let $f(x) = \tan^{-1} x$, which satisfies the given properties in the sufficient condition.

Note: For $\int_0^a \tan^{-1} x \, dx$:

$$\begin{array}{lll} \text{Let} & u = \tan^{-1} x & \rightarrow u' = \frac{1}{1+x^2} \\ & v' = 1 & \rightarrow v = x \end{array}$$

Then by the implication from part (i):

$$\begin{aligned} \int_0^a \tan^{-1} x \, dx + \int_0^b \tan y \, dy &\geq ab \quad \checkmark \\ \left(\left[x \tan^{-1} x \right]_0^a - \int_0^a \frac{x}{1+x^2} \, dy \right) + \int_0^b \frac{\sin y}{\cos y} \, dy &\geq ab \quad \checkmark \end{aligned}$$

$$a \tan^{-1} a - \frac{1}{2} \left[\ln(1+x^2) \right]_0^a - \left[\ln(\cos y) \right]_0^b \geq ab$$

(Note: $1+x^2 > 0$ and $\cos y > 0$ for $x \in [0, a]$ and $y \in [0, b]$)

$$a \tan^{-1} a - \frac{1}{2} \ln(1+a^2) - \ln(\cos b) \geq ab \quad \checkmark$$

$$\begin{aligned} \tan^{-1} a &\geq b + \frac{1}{a} \ln(\sqrt{1+a^2}) + \frac{1}{a} \ln(\cos b) \\ &= b + \frac{1}{a} \ln(\cos b \cdot \sqrt{1+a^2}) \end{aligned}$$

Let $a = \frac{1}{2}$ and $b = \frac{\pi}{6}$:

$$\begin{aligned} \tan^{-1} \left(\frac{1}{2} \right) &\geq \frac{\pi}{6} + 2 \ln \left(\frac{\sqrt{3}}{2} \cdot \sqrt{\frac{5}{4}} \right) \\ &= \frac{\pi}{6} + 2 \ln \left(\frac{\sqrt{15}}{16} \right) \\ \therefore \tan^{-1} \left(\frac{1}{2} \right) &\geq \frac{\pi}{6} + \ln \left(\frac{15}{16} \right) \quad \checkmark \end{aligned}$$

(b) i. (4 marks)

- ✓ [1] for the correct acceleration equation from the sum of forces
- ✓ [1] for the correct exit speed of the object
- ✓ [1] for the correct integral from $\frac{dv}{dt}$
- ✓ [1] for correctly showing the required result with justifications (resolving $\ln|f(x)|$)

Let Object A be the first object, and downwards be the positive direction of its motion.

$$\begin{aligned}\sum F_A &= mg - \frac{v^2}{45} \\ 2\ddot{x} &= 20 - \frac{v^2}{45} \\ \ddot{x}_A &= \frac{900 - v^2}{90} \quad \checkmark\end{aligned}\tag{1}$$

Let $v = V$ when $x = 25$:

$$\begin{aligned}v \frac{dv}{dx} &= \frac{900 - v^2}{90} \\ \int_0^V \frac{v}{900 - v^2} dv &= \int_0^{25} \frac{1}{90} dx \\ -\frac{1}{2} \left[\ln|900 - v^2| \right]_0^V &= \frac{1}{90} \left[x \right]_0^{25} \\ \ln \left(\frac{900 - V^2}{900} \right) &= -\frac{1}{45} (25 - 0) \quad (\text{since } \ddot{x}_A > 0, \quad 900 - v^2 > 0) \\ 900 - V^2 &= 900e^{-\frac{5}{9}} \\ V^2 &= 900 \left(1 - e^{-\frac{5}{9}} \right) \\ V &= 19.5862... \text{ ms}^{-1} \quad \checkmark \quad (\text{since } V > 0)\end{aligned}$$

(i.e. V is the velocity at which Object A exits the gaseous medium.)

From (1):

$$\begin{aligned}\frac{dv}{dt} &= \frac{(30 + v)(30 - v)}{90} \\ \int_0^v \frac{1}{(30 + v)(30 - v)} dv &= \int_0^t \frac{1}{90} dt \\ \int_0^v \frac{\frac{1}{60}}{30 + v} + \frac{\frac{1}{60}}{30 - v} dv &= \frac{1}{90} \left[t \right]_0^t \\ \left[\ln|30 + v| - \ln|30 - v| \right]_0^v &= \frac{2}{3} t \quad \checkmark \\ (\text{since } \ddot{x} > 0 \text{ and } v > 0, \quad (30 + v)(30 - v) > 0 \text{ so } 30 - v > 0) \\ \ln \left(\frac{30 + v}{30 - v} \right) - \ln \left(\frac{30}{30} \right) &= \frac{2}{3} t \\ t_A &= \frac{3}{2} \ln \left(\frac{30 + v}{30 - v} \right)\end{aligned}$$

When $v = V \approx 19.5862...$:

$$t = 2.3408...$$

Hence Object A exits the gaseous medium approximately 2.34 seconds after it was dropped. $\checkmark$

Alternate method:

Let Object A be the first object, and downwards be the positive direction of its motion.

$$\begin{aligned}\sum F_A &= mg - \frac{v^2}{45} \\ 2\ddot{x} &= 20 - \frac{v^2}{45} \\ \ddot{x}_A &= \frac{900 - v^2}{90} \quad \checkmark\end{aligned}\tag{1}$$

$$\begin{aligned}\frac{dv}{dt} &= \frac{(30+v)(30-v)}{90} \\ \int_0^v \frac{1}{(30+v)(30-v)} dv &= \int_0^t \frac{1}{90} dt \\ \int_0^v \frac{\frac{1}{60}}{30+v} + \frac{\frac{1}{60}}{30-v} dv &= \frac{1}{90} [t]_0^t \\ \left[\ln|30+v| - \ln|30-v| \right]_0^v &= \frac{2}{3}t \quad \checkmark \\ (\text{since } \ddot{x} > 0 \text{ and } v > 0, \quad (30+v)(30-v) > 0 \text{ so } 30-v > 0) \\ \ln\left(\frac{30+v}{30-v}\right) - \ln\left(\frac{30}{30}\right) &= \frac{2}{3}t \\ \frac{30+v}{30-v} &= e^{\frac{2}{3}t} \\ (1 + e^{\frac{2}{3}t})v &= 30(e^{\frac{2}{3}t} - 1) \\ v &= \frac{30(e^{\frac{2}{3}t} - 1)}{e^{\frac{2}{3}t} + 1}\end{aligned}$$

$$\begin{aligned}[x]_0^x &= \int_0^t \frac{e^{\frac{1}{3}t} - e^{-\frac{1}{3}t}}{e^{\frac{1}{3}t} + e^{-\frac{1}{3}t}} dt \\ x &= 90 \left[\ln \left| e^{\frac{1}{3}t} + e^{-\frac{1}{3}t} \right| \right]_0^t \\ &= 90 \left(\ln \left(e^{\frac{1}{3}t} + e^{-\frac{1}{3}t} \right) - \ln 2 \right) \quad \checkmark \\ &\quad (\text{since } e^{\frac{1}{3}t} > 0 \text{ and } e^{-\frac{1}{3}t} > 0)\end{aligned}$$

When $t = 2.34$,

$$\begin{aligned}x &= 90 \left(\ln \left(e^{\frac{2.34}{3}} + e^{-\frac{2.34}{3}t} \right) - \ln 2 \right) \\ &= 24.98 \dots \\ &\approx 25\end{aligned}$$

Hence Object A exits the gaseous medium approximately 2.34 seconds after it was dropped. $\checkmark$

ii. (3 marks)

- ✓ [1] for the correct height of Object B when Object A exits the gas
- ✓ [1] for correctly showing that Object B is moving upwards when Object A exits the gas
- ✓ [1] for the correct conclusion with sufficient justifications

Let Object B be the second object, and upwards be the positive direction of its motion.

$$\begin{aligned}
 \sum F_B &= -mg - \frac{v}{10} \\
 2\ddot{x} &= -20 - \frac{v}{10} \\
 \ddot{x}_B &= \frac{-(200 + v)}{20} \\
 \frac{dv}{dt} &= -\frac{1}{20}(200 + v) \\
 \int_{40}^v \frac{1}{200 + v} dv &= \int_0^t -\frac{1}{20} dt \\
 \left[\ln |200 + v| \right]_{40}^v &= -\frac{1}{20} \left[t \right]_0^t \\
 \ln \left(\frac{200 + v}{240} \right) &= -\frac{1}{20} t && \text{(since } \ddot{x}_B < 0, \quad 200 + v > 0) \\
 200 + v &= 240e^{-\frac{1}{20}t} \\
 v_B &= 240e^{-\frac{1}{20}t} - 200 && (2)
 \end{aligned}$$

$$\begin{aligned}
 \frac{dx}{dt} &= 240e^{-\frac{1}{20}t} \\
 \left[x \right]_0^x &= \left[-4800e^{-\frac{1}{20}t} - 200t \right]_0^t \\
 x &= -4800e^{-\frac{1}{20}t} - 200t + 4800 \\
 x_B &= 4800 \left(1 - e^{-\frac{1}{20}t} \right) - 200t
 \end{aligned}$$

From part (i), when $t = 2.3408\dots$:

$$\begin{aligned}
 x_B &= 62.00\dots \quad \checkmark \\
 v_B &= 13.49\dots > 0
 \end{aligned}$$

i.e. when Object A exits the gaseous medium, Object B is 62 m above the ground and still moving upwards (has yet to reach its peak or fall back down). $\checkmark$

The gaseous medium has a depth of 25 m, below which is 65 m of air before the ground. Thus Object B cannot enter the gaseous medium before Object A exits it.

Hence the two objects will collide in the air. $\checkmark$